
Encryption and Decryption

1. Encryption

a. Caesar Cipher

Text: CHASE YOUR DREAMS

Formula:

Encryption: $C = E(P + n) \bmod 26$

Decryption: $P = D(C - n) \bmod 26$

Key: $n = 3$

Answer: FKDVH BRXU GUHDPV

The Caesar cipher shifts each letter three positions forward in the alphabet while keeping the spaces unchanged.

b. Shift Cipher

Text: CHASE YOUR DREAMS

Formula:

Encryption: $C = E(P + n) \bmod 26$

Decryption: $P = D(C - n) \bmod 26$

Key: $n = 7$

Answer: JOHZL FVBY KYLHTZ

Here, each letter is shifted seven positions forward according to the given key.

c. ROT13

Text: CHASE YOUR DREAMS

Formula:

Encryption: $C = E(P + n) \bmod 26$

Decryption: $P = D(C - n) \bmod 26$

Key: $n = 13$

Answer: PUNFR LBHE QERNZF

ROT13 replaces each letter with the letter 13 positions away from it in the alphabet.

2. Decryption

a. Caesar Cipher

Text: WRWDOOB VHFUHW PHVVDJH

Formula:

Encryption: $C = E(P + n) \bmod 26$

Decryption: $P = D(C - n) \bmod 26$

Key: $n = 3$

Answer: TOTALLY SECRET MESSAGE

Each letter is shifted three positions backward to recover the original message.

b. Shift Cipher

Text: SJI IBTW YNRJ

Formula:

Encryption: $C = E(P + n) \bmod 26$

Decryption: $P = D(C - n) \bmod 26$

Key: $n = 5$

Answer: NED DWOR TIME

Using a backward shift of five gives the above result. The given ciphertext appears to contain an error because it does not produce a meaningful English sentence.

c. ROT13

Text: FRAQ ZBER PBSSRR

Formula:

Encryption: $C = E(P + n) \bmod 26$

Decryption: $P = D(C - n) \bmod 26$

Key: $n = 13$

Answer: SEND MORE COFFEE

Each letter is shifted 13 positions backward, which reverses the ROT13 transformation and gives the original message.
